

Fiber Optic Network Design

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Outline

- **FWM: definition & origin**
- **FWM in fiber, effect on WDM**
- **How to reduce it & simulation**
 - **Solutions**
 - **Polarization allocation**
 - **Unequal channel spacing**
 - **Dispersion management**
 - **Increase effective area of fiber**

Four-Wave Mixing (FWM)

Definition:

Nonlinear effect that occurs in nonlinear optical materials such as photonic switch, optical fiber cable, etc. This interaction between waves leads to interaction between channels

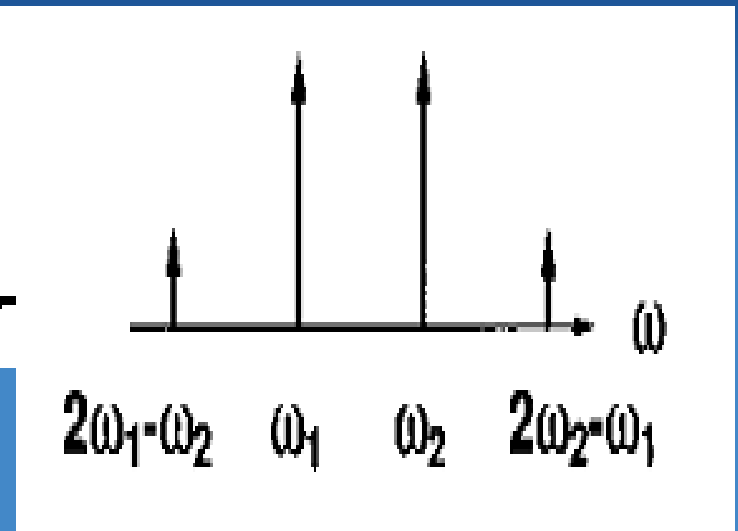
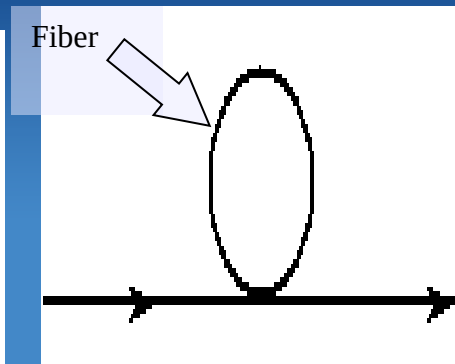
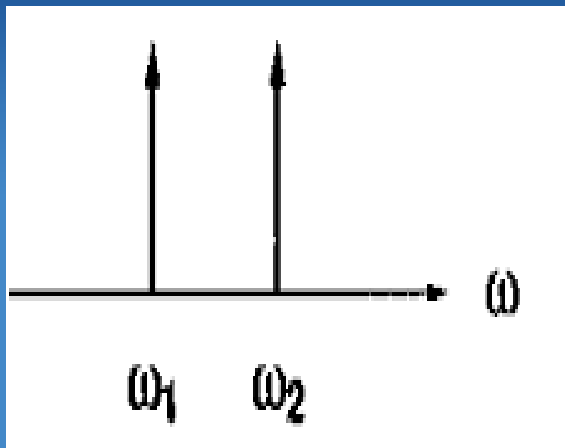
The effects of FWM

generates one or more new channels (or harmonics). For instance, when three waves at frequencies f_i , f_j , and f_k traverse a fiber they generate another signal located at

$$f_{ijk} = f_i + f_j - f_k$$

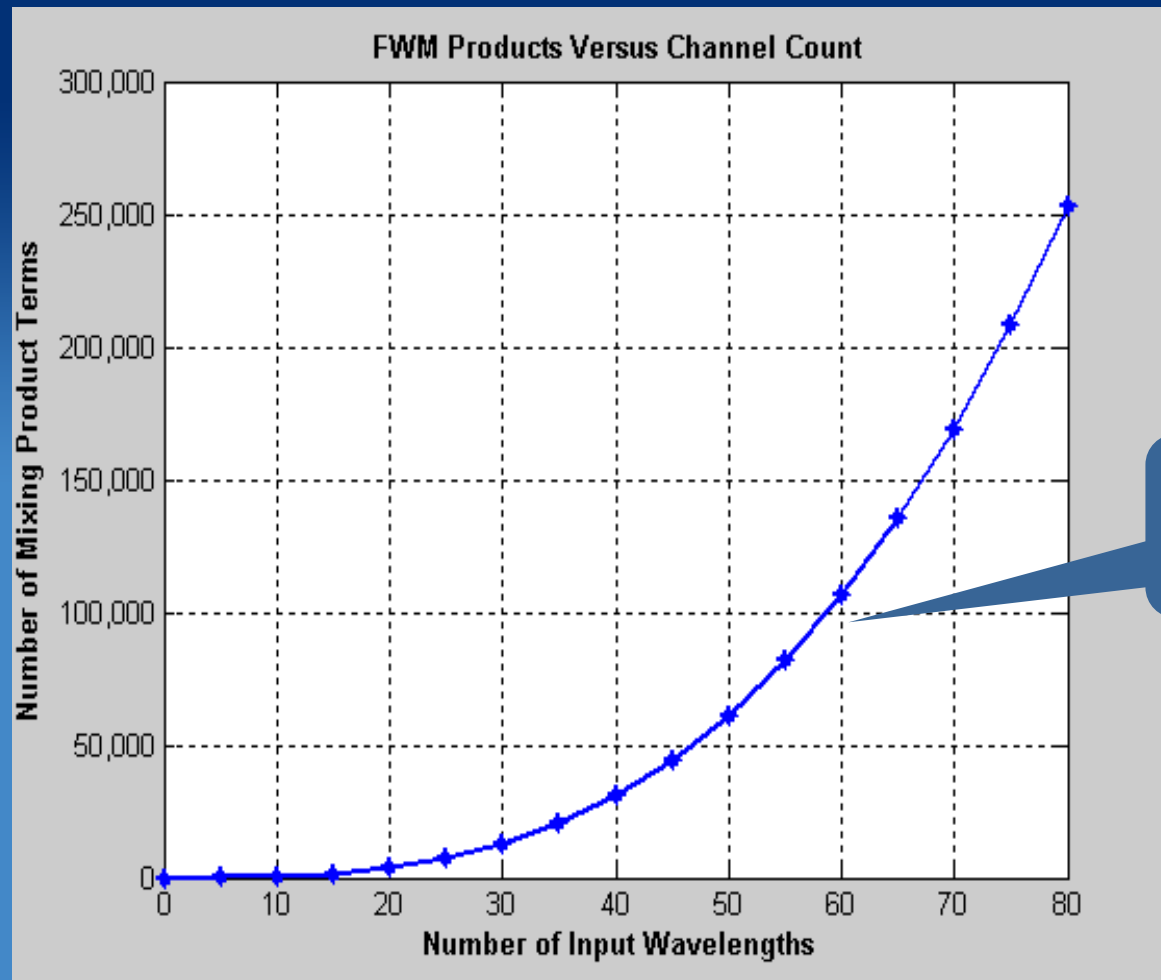
When $f_i = f_j$, FWM is degenerated.

Example : The input is two signals located at ω_1 and ω_2 traverse a fiber of length (L) and the output is four different signals located at ω_1 , ω_2 , $2\omega_1 - \omega_2$, and $2\omega_2 - \omega_1$.

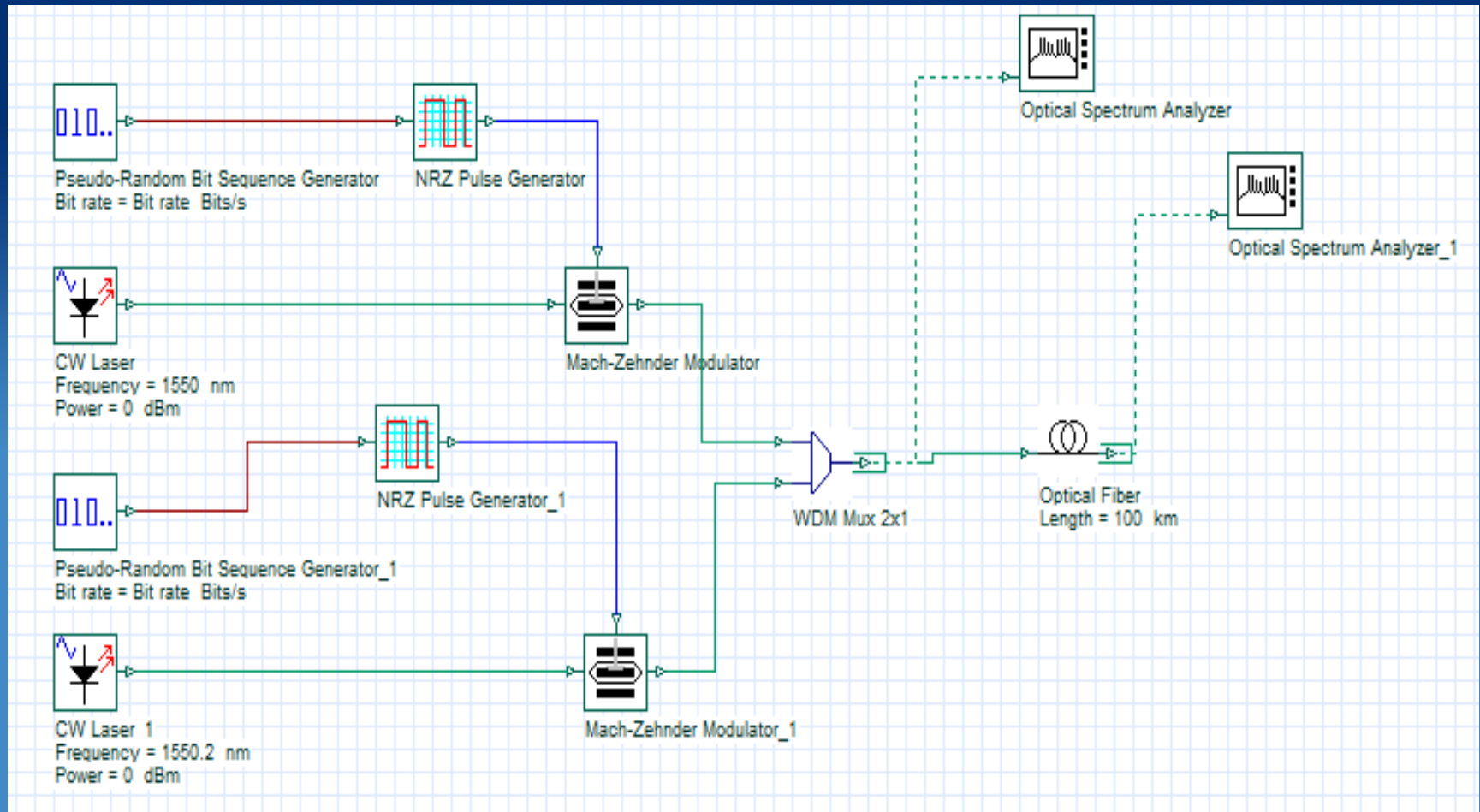


In general the number of newly generated signals is given by

$$M = 2C_N^2 + 3C_N^3 = 2 \frac{N!}{2!} + 3 \frac{N!}{3!} = \frac{N^2(N-1)}{2}$$

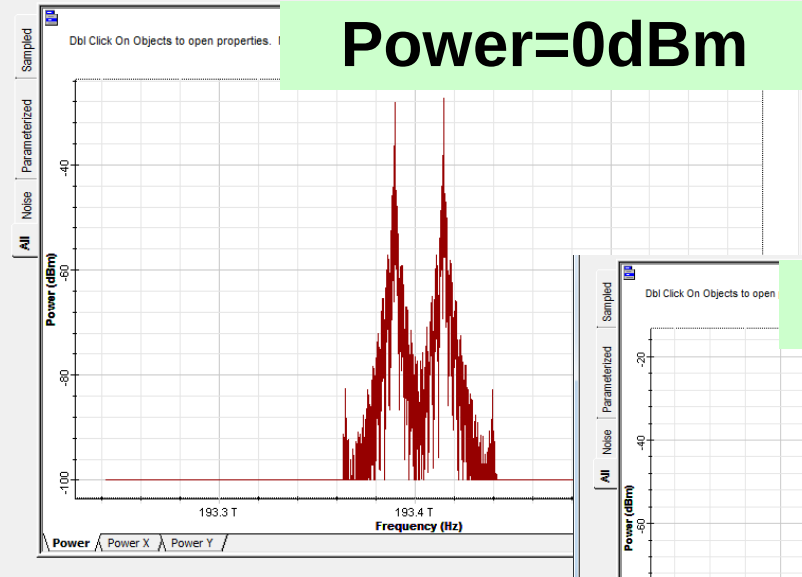


System Used for Generating FWM

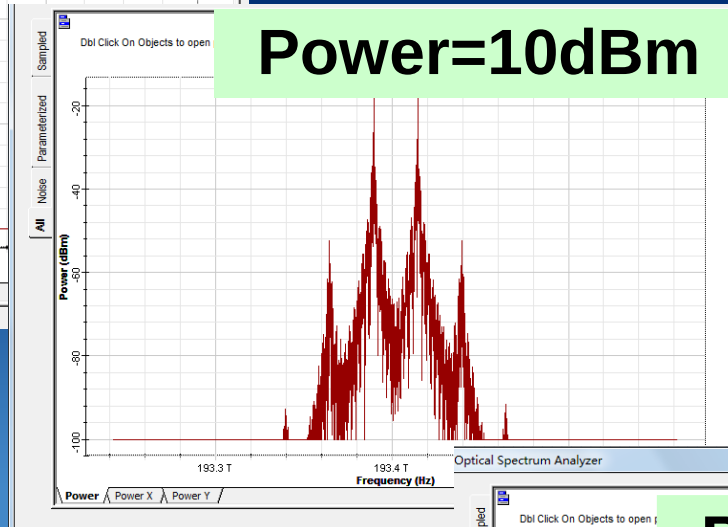


Effect generated by Optisystems

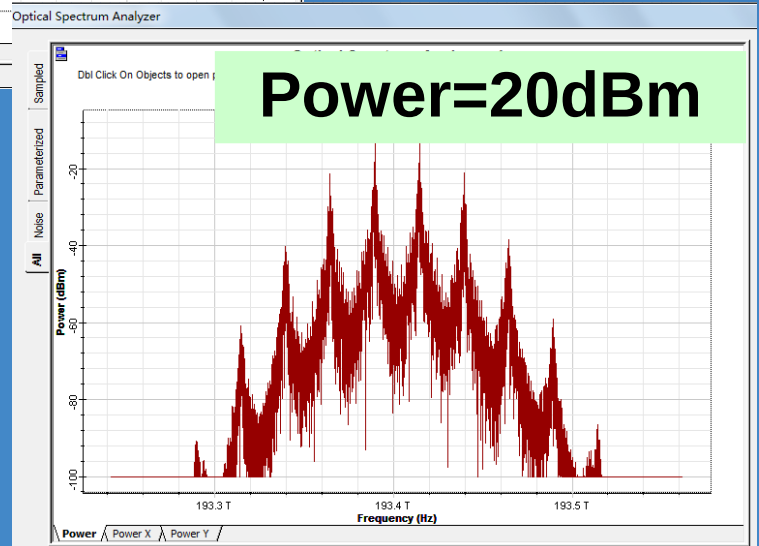
Power=0dBm



Power=10dBm



Power=20dBm



Origin of FWM

- Third order nonlinearity

$$P_{NL} = \varepsilon_0 \chi^{(3)} : EEE$$

- Energy Conservation

$$\omega_4 = \omega_1 + \omega_2 - \omega_3$$

- Momentum Conservation—phase matching

$$\beta_4 = \beta_1 + \beta_2 - \beta_3$$

P_{NL} is the electric polarization (rather than power).

Origin of FWM

Amplitude:

$$\frac{dA_F}{dz} = -\frac{\alpha}{2} A_F + d_F \gamma A_i A_j A_k^* \exp(-i\Delta k z)$$

Neglecting the SPM
and XPM items

Power:

$$P_F = |A_F(L)|^2 = \eta_F (d_F \gamma L)^2 P_i P_j P_k e^{-\alpha L}$$

A_F is the amplitude of the FWM component at ω_F .

$A_m(z) = A_m(0)\exp(-\alpha z/2)$ for $m = i, j, k$; $P_m = |A_m(0)|^2$ at $z = 0$.

$d_F = 2 - \delta_{ij}$ is the degeneracy factor; $d_F = 1$ when $i = j$ and $d_F = 2$ when $i \neq j$.

Nonlinear parameter:

$$\gamma = \frac{2\pi n_2}{\lambda A_{eff}} = \frac{3\pi \operatorname{Re}(\chi^{(3)})}{4n\lambda A_{eff}}$$

A_{eff} is the effective fiber cross-sectional area.

$\chi^{(3)}$ is the third-order nonlinear susceptibility.

Origin of FWM ---- Phase Matching

Phase mismatch:

$$\Delta k = \beta_F + \beta_k - \beta_i - \beta_j \approx \beta_2 (\omega_i - \omega_k) (\omega_j - \omega_k)$$

FWM efficiency:

$$\eta_F = \left| \frac{1 - \exp[-(\alpha + i\Delta k)L]}{(\alpha + i\Delta k)L} \right|^2$$

$$\Delta k \ll 1, \eta_F \rightarrow 1$$

$$\Delta k \gg 1, \eta_F \rightarrow 0$$

Coherence length:

$$L_{coh} = \frac{2\pi}{|\Delta k|}$$

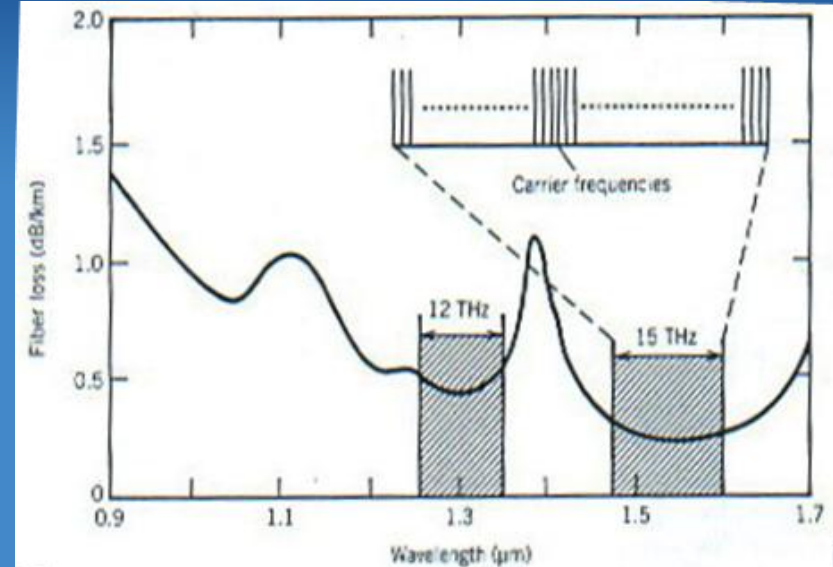
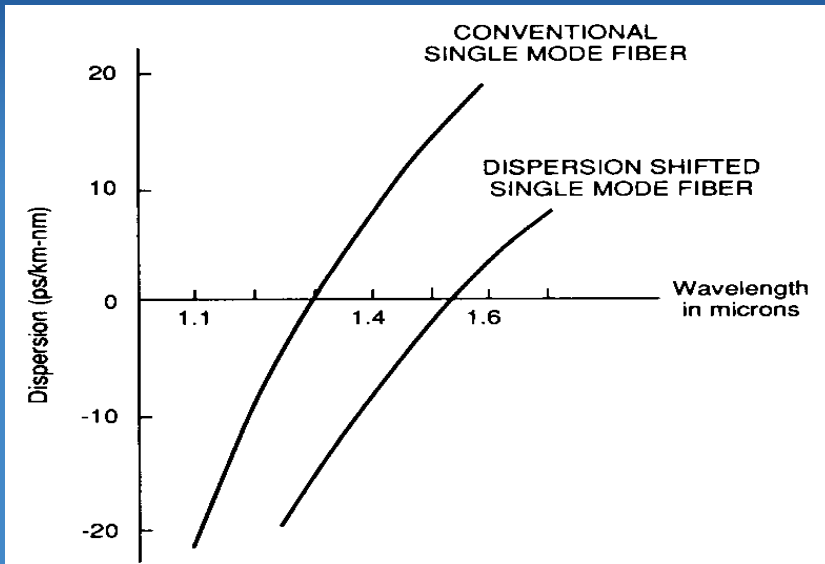
$$L < L_{coh}$$

Effects on WDM Systems

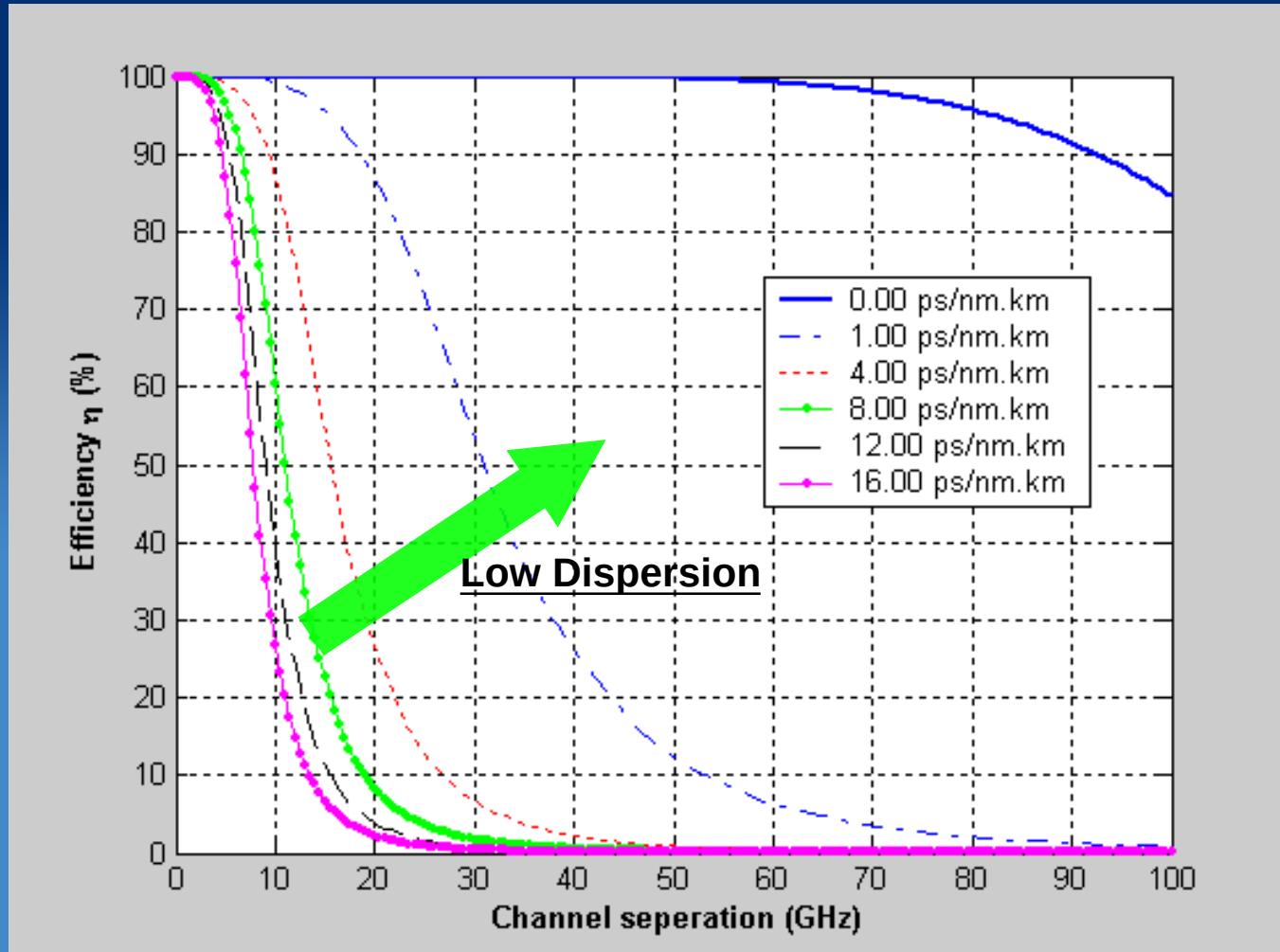
Phase Match Conditions

$$\Delta k = \beta_F + \beta_k - \beta_i - \beta_j \approx \beta_2 (\omega_i - \omega_k)(\omega_j - \omega_k)$$

- When fiber dispersion and channel spacing are small enough

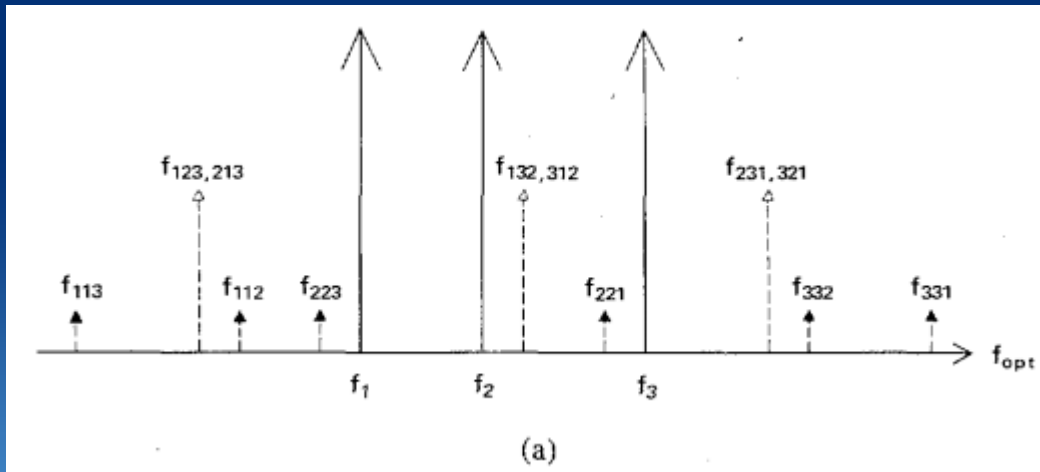


The efficiency curve versus the channel separation in (GHz) & dispersion



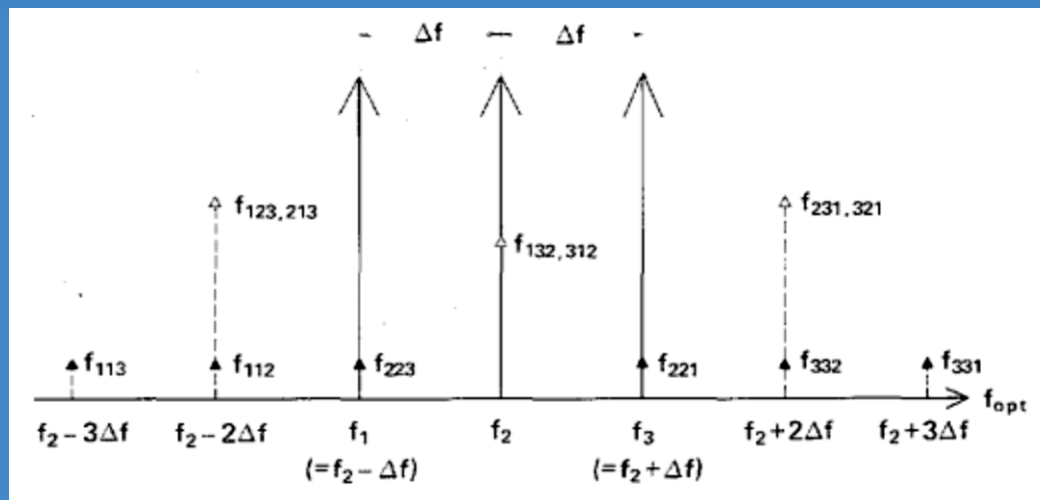
Effects on WDM Systems

- Nonlinear Crosstalk



N channels $f_{ijk} = f_i + f_j - f_k$

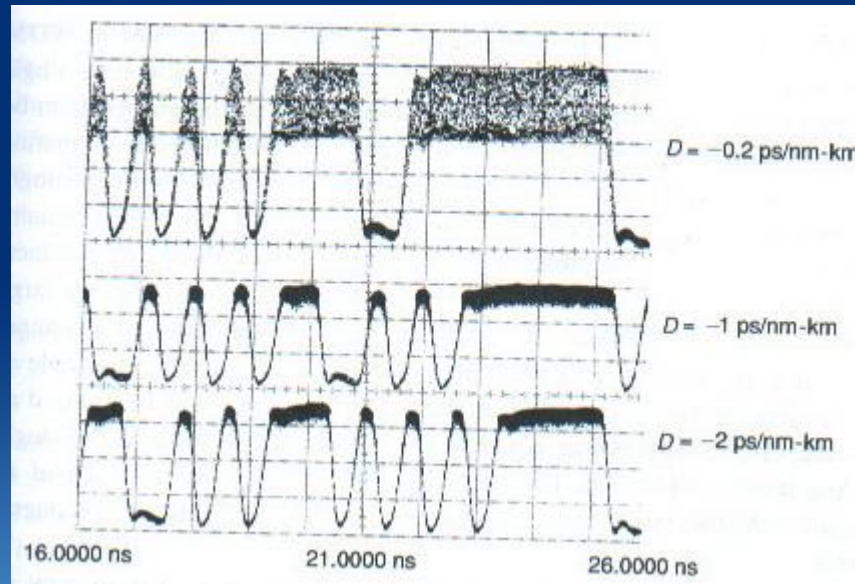
$\frac{1}{2}N^2(N-1)$ mixing products



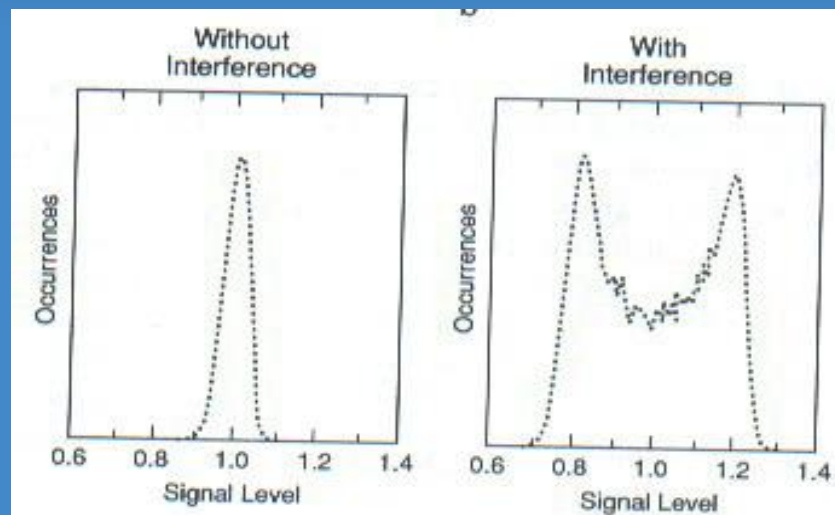
equally spacing,
product terms fall
at channel frequency
Coherent interference

Effects on WDM Systems

- Bit pattern



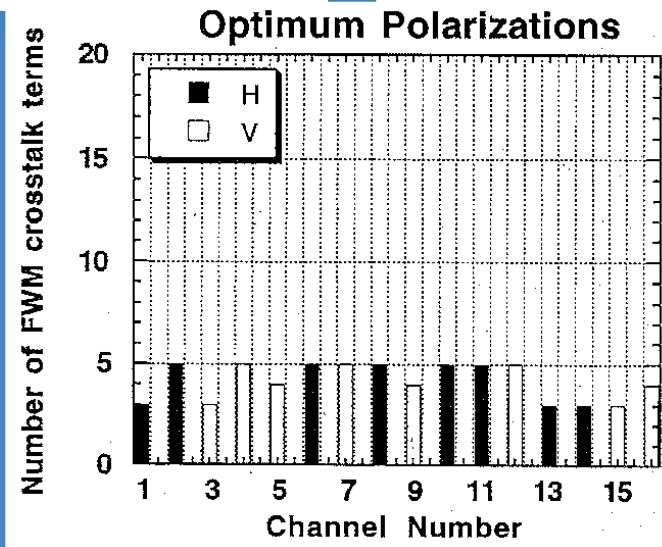
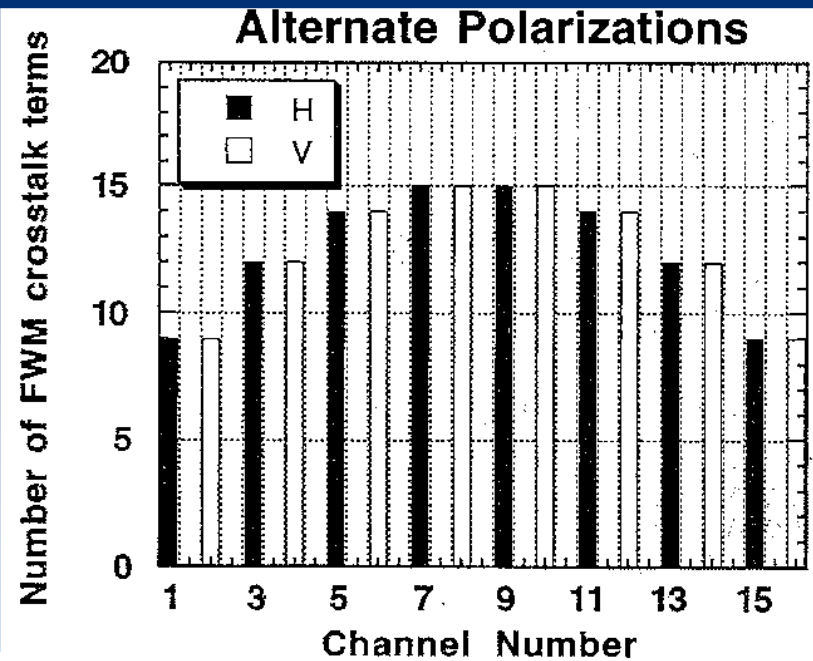
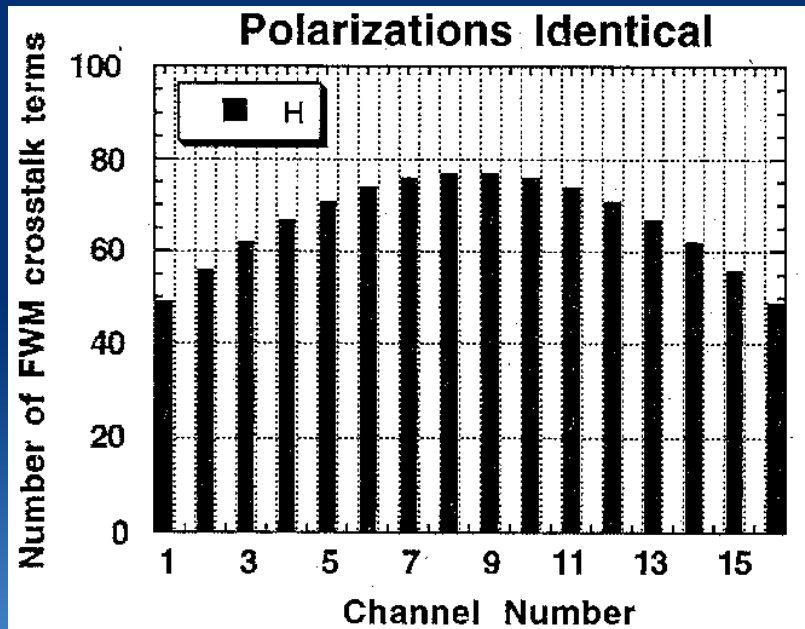
- Power Depletion



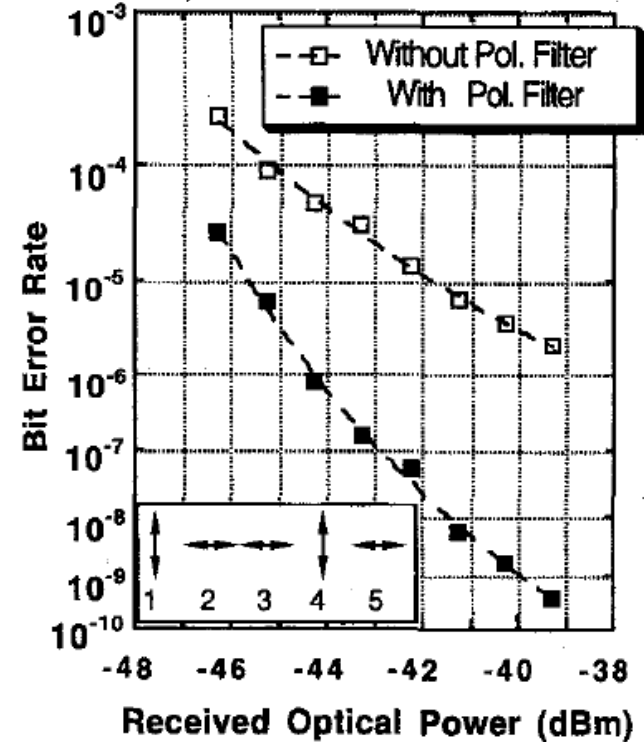
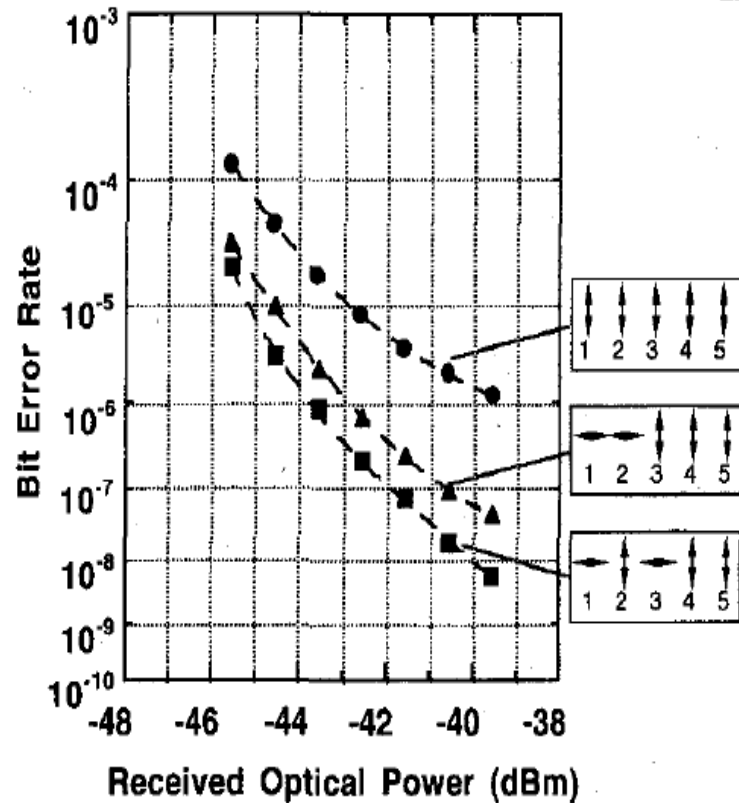
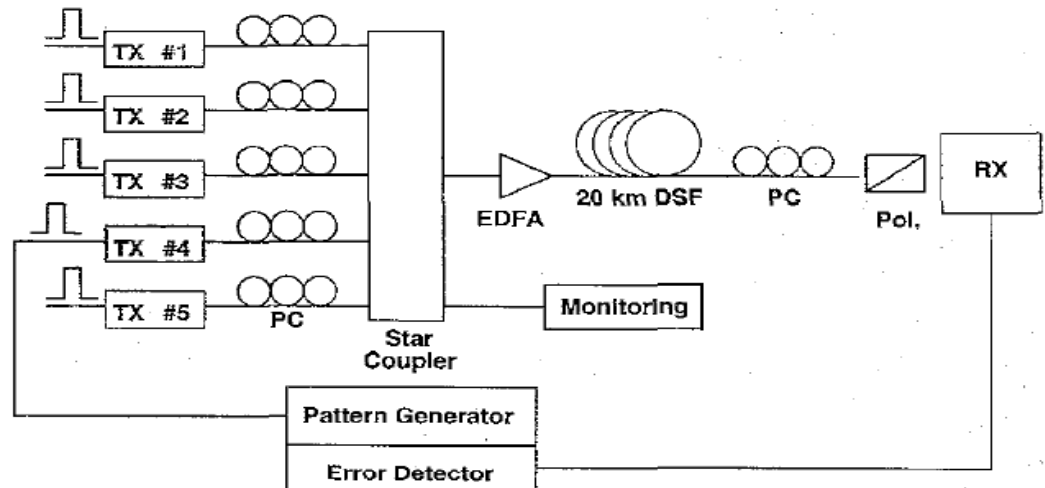
Possible Solutions to reduce FWM

- Polarization allocation
- Unequal channel spacing
- Dispersion management
- Increase effective area of fiber

Polarization Allocation



Polarization Allocation

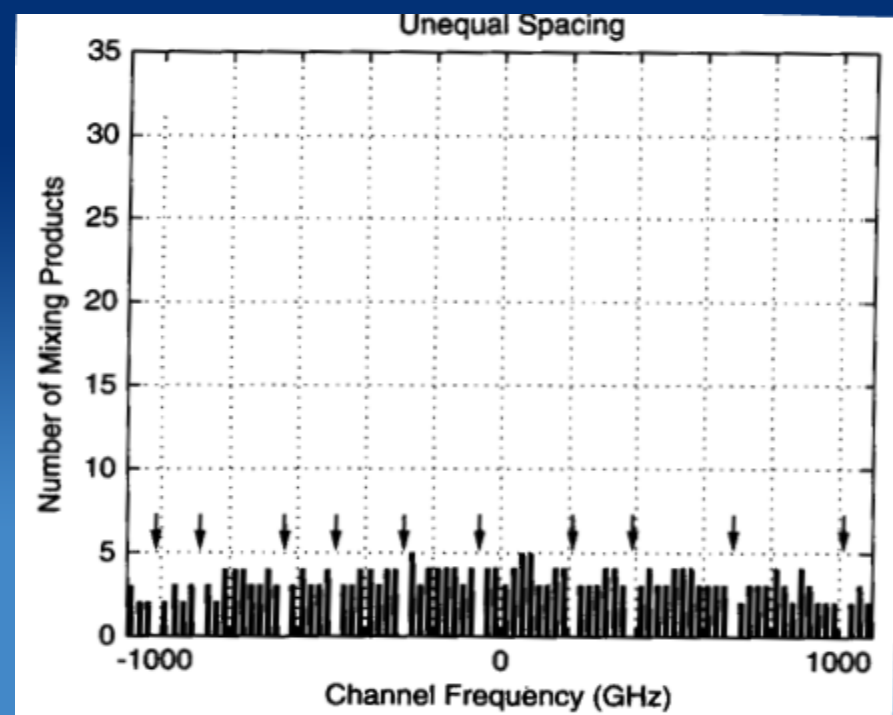
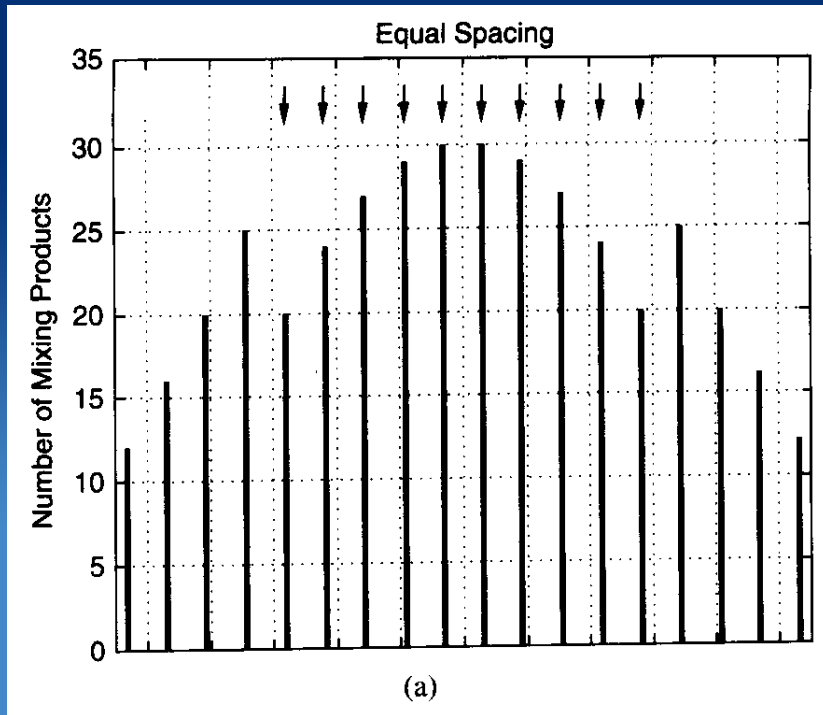


Unequal Channel Spacing

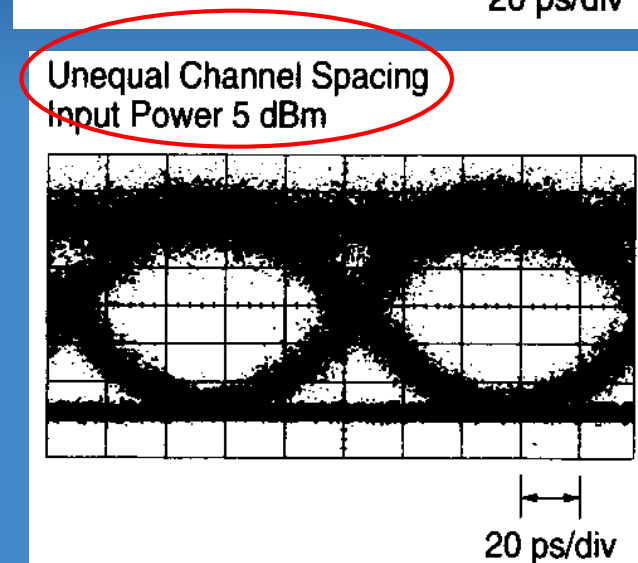
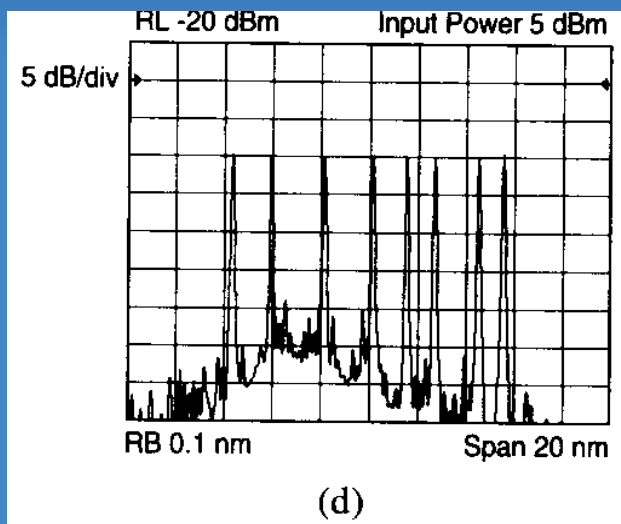
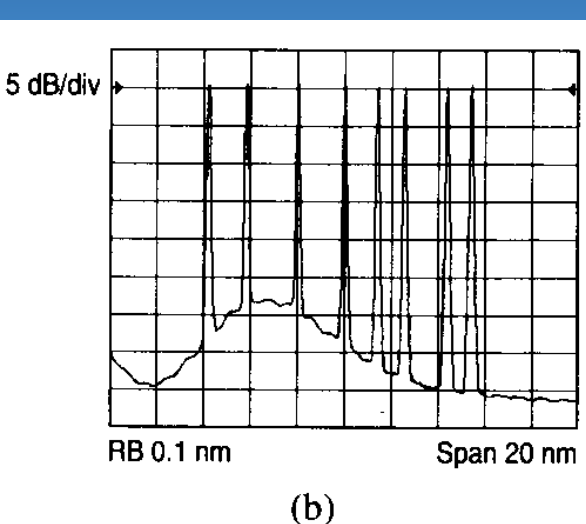
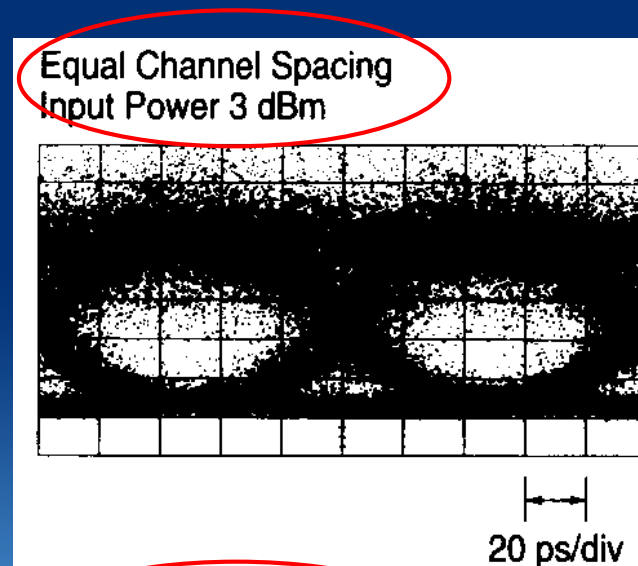
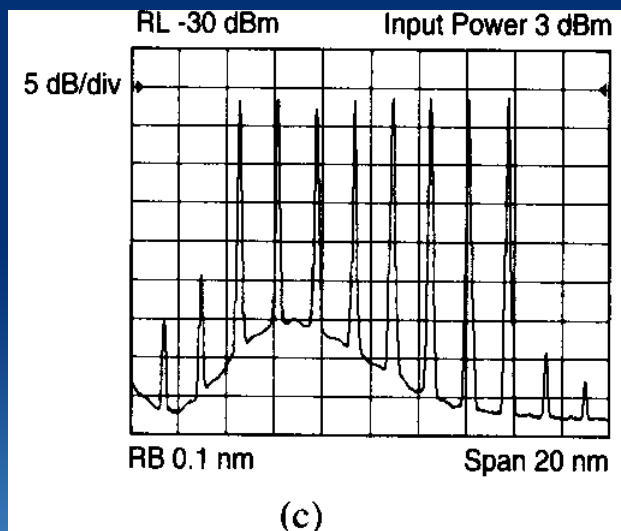
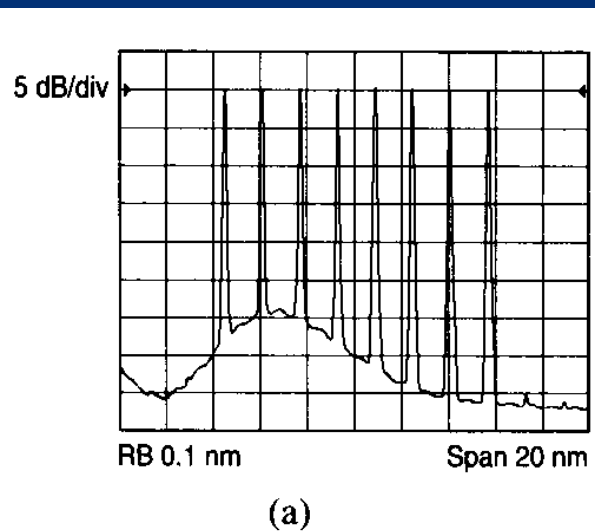
Equal Spacing



Unequal Spacing

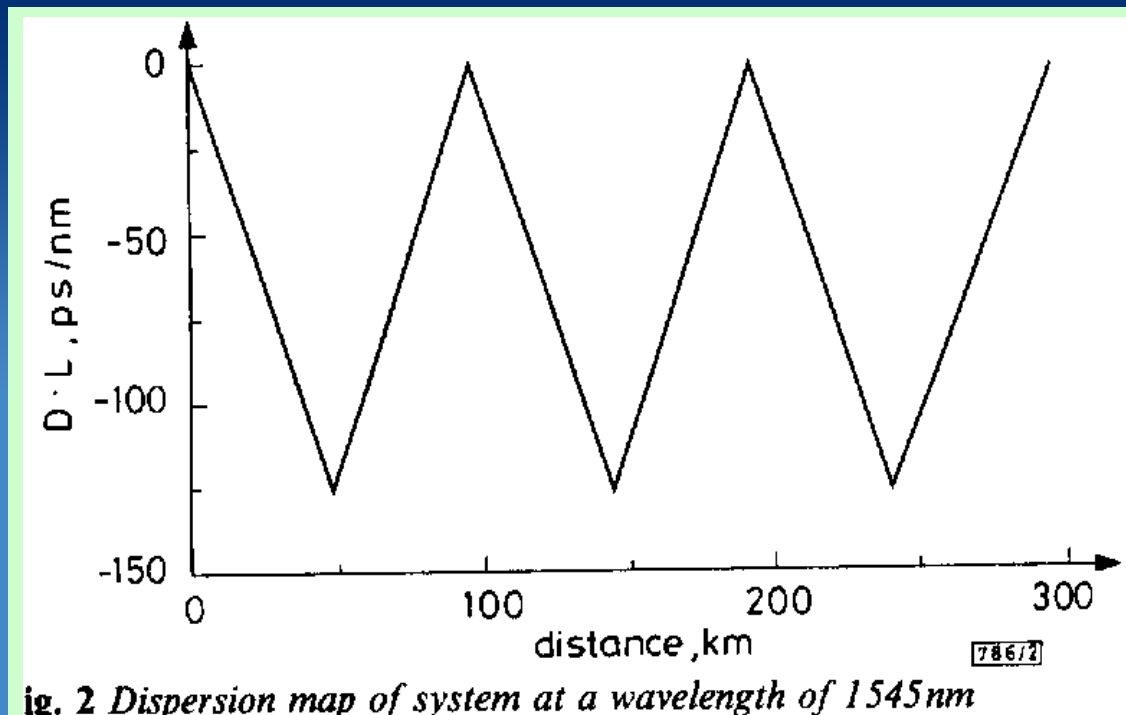


Unequal Channel Spacing



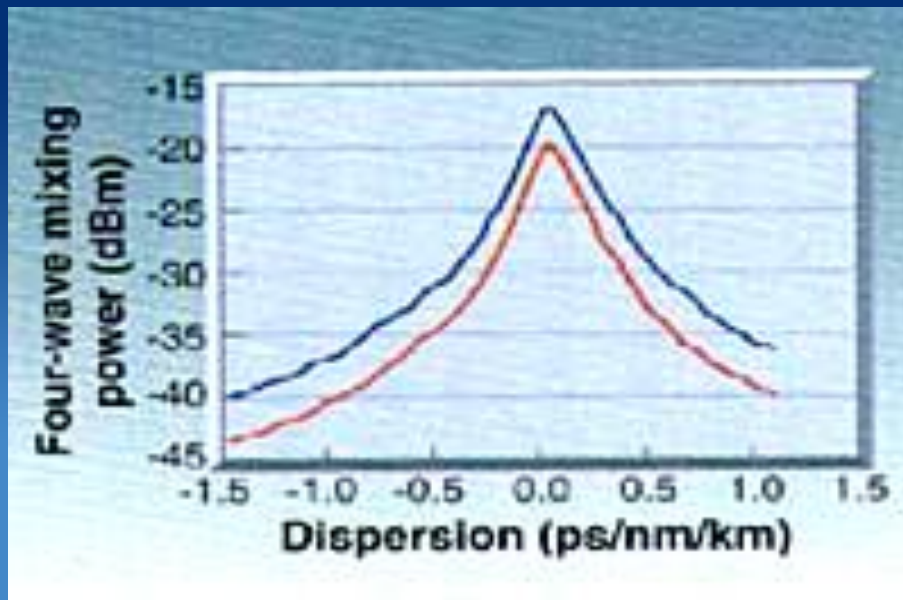
Dispersion-Management

- No dispersion-zero point
- Total accumulated dispersion near zero



New Type of Fibers

- Large effective area fiber
- Nonzero dispersion shifted fiber (NZ-DSF)



Large-effective-area fiber (red) suppressed four-wave mixing crosstalk to -22.5 dB, compared to -19.0 dB for NZ-DSF fiber (blue).

Today's Task

- Set up a WDM system to evaluate FWM effect
- Set all laser power = 0 dbm
 - Increase fiber effective area to reduce FWM and increase BER
 - Try different dispersion settings
 - Try different polarization settings to reduce BER
 - Try unequal spacing to reduce BER